

Dose-response efficacy of a proprietary probiotic formula of Lactobacillus acidophilus CL1285 and Lactobacillus casei LBC80R for antibiotic-associated...



OBJECTIVES: Standard therapies for antibiotic-associated diarrhea (AAD) and Clostridium difficile-associated diarrhea (CDAD) have limited efficacy. Probiotic prophylaxis is a promising alternative for reduction of AAD and CDAD incidence. METHODS: In this single-center, randomized, double-blind, placebo-controlled dose-ranging study, we randomized 255 adult inpatients to one of three groups: two probiotic capsules per day (Pro-2, n=86), one probiotic capsule and one placebo capsule per day (Pro-1, n=85), or two placebo capsules per day (n=84). Each probiotic capsule contained 50 billion c.f.u. of live organisms (Lactobacillus acidophilus CL1285 +Lactobacillus casei LBC80R Bio-K+ CL1285). Probiotic prophylaxis began within 36 h of initial antibiotic administration, continued for 5 days after the last antibiotic dose, and patients were followed for an additional 21 days. RESULTS: Pro-2 (15.5%) had a lower AAD incidence vs. Pro-1 (28.2%). Each probiotic group had a lower AAD incidence vs. placebo (44.1%). In patients who acquired AAD, Pro-2 (2.8 days) and Pro-1 (4.1 days) had shorter symptom duration vs. placebo (6.4 days). Similarly, Pro-2 (1.2%) had a lower CDAD incidence vs. Pro-1 (9.4%). Each treatment group had a lower CDAD incidence vs. placebo (23.8%). Gastrointestinal symptoms were less common in the treatment groups vs. placebo and in Pro-2 vs. Pro-1. CONCLUSIONS: The proprietary probiotic blend used in this study was well tolerated and effective for reducing risk of AAD and, in particular, CDAD in hospitalized patients on antibiotics. A dose-ranging effect was shown with 100 billion c.f.u., yielding superior outcomes and fewer gastrointestinal events compared to 50 billion c.f.u. (ClinicalTrials.gov number NCT00958308).